Question 1

M1L3

From the lecture slides:

- Can simulate almost anything.
  - Customer-based systems like Manufacturing Processes, Supply Chains, Health Systems.

Advantages ☺

- Can study models too complicated for analytical / numerical treatment.

Question 2

M2L2

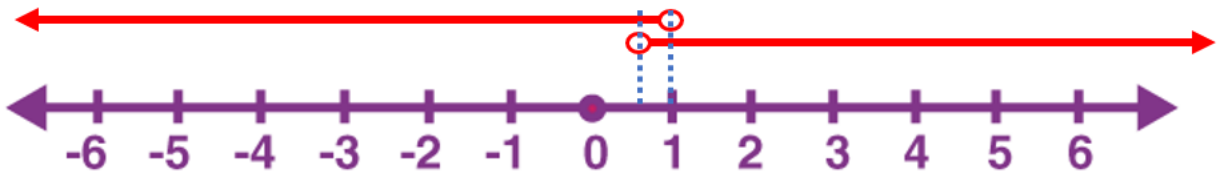
Since it is multiple choice, you could use trial and error and substitute each multiple choice answer to see if it is a solution.

Question 3

M2L5

Use a number line to see that $P(X < 1 \cap X > 1/2) = P(1/2 < X < 1)$

The two inequalities overlap between ½ and 1.



Another way to think about it is that the conditional probability limits us to the part of the number line greater than ½. So now we are only looking at values between ½ and 2. So when we ask for a probability that is less than 1 given the condition, then the numerator must be between ½ and 1 while we divide by the probability of the new sample space of ½ to 2.

Question 4

M2L12

This is a direct result of using the PMF of the Geom(0.6) distribution.

$X \sim \text{Geometric}(p)$ is the number of Bern(p) trials until a success occurs. For example, "FFFS" implies that $X = 4$.

$$f(x) = q^{x-1}p, \quad x = 1, 2, \ldots.$$

Question 5

M2L7 M2L12

This is a direct result of using LOTUS with $h(X) = e^X$ and the PMF of a Bernoulli random variable.

$X \sim \text{Bernoulli}(p)$.

$$f(x) = \begin{cases} p & \text{if } x = 1 \\ 1 - p \ (= q) & \text{if } x = 0 \end{cases}$$

Def/Thm (the "Law of the Unconscious Statistician" or "LOTUS"): Suppose that $h(X)$ is some function of the RV X. Then

$$\mathrm{E}[h(X)] = \begin{cases} \sum_x h(x) f(x) & \text{if } X \text{ is disc} \\ \int_{\mathbb{R}} h(x) f(x)\, dx & \text{if } X \text{ is cts} \end{cases} = \int_{\mathbb{R}} h(x)\, dF(x).$$

Question 6

M2L7

This is a direct result using $E[aX + b] = aE[X] + b$. The variance isn't needed because all we are concerned with is an expected value.

Question 7

M2L12

This is a direct result of the expected value of a Geom(1/10) random variable.

You can also think of this intuitively – if there are 10 sides to the dice then we would expect to have to roll it 10 times in order to observe a particular number.

Question 8

M2L11

We are given $\mathrm{Var}(X) = \mathrm{Var}(Y) = 3$ and $\mathrm{Cov}(X, Y) = 0$

Once you see that the covariance is zero this should immediately imply that the correlation is 0.

Question 9

M2L11

Recall

$$\mathrm{Var}(X - Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) - 2\mathrm{Cov}(X, Y)$$
$$\mathrm{Var}(aX + b) = a^2\mathrm{Var}(X)$$
$$\mathrm{Cov}(aX, bY) = ab\mathrm{Cov}(X, Y)$$

Do not get confused by the negative sign here. The formula in the first line already accounts for this. If it helps, you can think of it as $\mathrm{Var}(X + (-Y))$

$$\mathrm{Var}(X+(-Y)) = \mathrm{Var}(X)+(-1)^2\mathrm{Var}(Y)+2(1)(-1)\mathrm{Cov}(\mathrm{X,Y})=\mathrm{Var}(\mathrm{X})+\mathrm{Var}(\mathrm{Y})\text{-}2\mathrm{Cov}(\mathrm{X,Y})$$

$$\mathrm{Var}(X - 2Y) = \mathrm{Var}(X) + \mathrm{Var}(-2Y) + 2\mathrm{Cov}(X, -2Y)$$

$$= \mathrm{Var}(X) + 4\mathrm{Var}(Y) - 4\mathrm{Cov}(X, Y)$$

$$= 10 + 4(10) - 4(6) = 10 + 40 - 24 = 50 - 24 = 26$$

Question 10

M2L12

Review Sections 4.2.2.1 in Goldsman, D. & Goldsman, P. (2020). A First Course in Probability and Statistics. Lulu. [Download link]

It can be shown that the interarrival times of a Poisson process are all independent and identically distributed exponential random variables with parameter λ.

Theorem: Let X be the amount of time until the *first arrival* of a Poisson process with rate λ. Then $X \sim \text{Exp}(\lambda)$.

Proof: The cdf of $F(x)$ is

$$\begin{aligned} F(x) &= \text{P}(X \leq x) \\ &= 1 - \text{P}(\text{no arrivals in } [0, x]) \\ &= 1 - \frac{e^{-\lambda x}(\lambda x)^0}{0!} \quad (\text{the number of arrivals in } [0, x] \text{ is Pois}(\lambda x)) \\ &= 1 - e^{-\lambda x}. \quad \square \end{aligned}$$

Theorem: Amazingly, it can be shown (after a lot of work) that the interarrival times of a Poisson process are *all* iid Exp(λ)! See for yourself when you take an advanced stochastic processes course.

Since we have a Poisson process, the distribution of the interarrival time between the 1st and 2nd arrivals is $\text{Exp}(\lambda = 1/2)$

Let the random variable $X \sim \text{Exp}(1/2)$

Use the CDF of the exponential distribution:

$$P(X < 1) = 1 - e^{-\frac{1}{2}(1)}$$

Using properties of exponents:

$$= 1 - \frac{1}{e^{1/2}} = 1 - \frac{1}{\sqrt{e}}$$

Question 11

M2L12

If you have a calculator that computes probabilities for Normal distributions with arbitrary mean and variance/standard deviation, you can do the calculation directly.

Otherwise, the random variable must be standardized and the standard Normal table should be used.

$$P(X > 0) = P\left(\frac{X-\mu}{\sigma} \geq \frac{0-2}{\sqrt{4}}\right) = P(Z > -1)$$

Review the document [How to use the Normal/t/χ^2 Tables]

From that document, we know that
$P(Z > -1) = 1 - P(Z \leq -1) = 1 - [1 - P(Z \leq 1)] = P(Z \leq 1)$ (you can also use the fact that the Normal distribution is symmetric to arrive at the same conclusion).

$$P(Z \leq 1) = 0.8413$$

2

Standard Normal Table

1 3

Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	[illegible]	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	[illegible]	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	[illegible]	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9031	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9924	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9958	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986

Question 12

M2L12

Recall from the lecture slides the following:

If $X_1, X_2, \ldots, X_n \overset{\text{iid}}{\sim} \text{Exp}(\lambda)$, then $Y \equiv \sum_{i=1}^{n} X_i \sim \text{Gamma}(n, \lambda)$. The $\text{Gamma}(n, \lambda)$ is also called the $\text{Erlang}_n(\lambda)$. It has cdf

$$F_Y(y) = 1 - e^{-\lambda y} \sum_{j=0}^{n-1} \frac{(\lambda y)^j}{j!}, \quad y \geq 0.$$

Since X and Y are iid exponential with $\lambda = 1$, the random variable $Z = X + Y$ is an Erlang random variable with $n = 2$ and $\lambda = 1$

The CDF of Z is

$$F_Z(z) = P(Z \leq z) = 1 - e^{-\lambda z} \sum_{j=0}^{n-1} \frac{(\lambda z)^j}{j!}$$

$$P(Z \leq 2) = 1 - e^{-2} \sum_{j=0}^{1} \frac{(2)^j}{j!} = 1 - e^{-2} \left[\frac{2^0}{0!} + \frac{2^1}{1!} \right] = 1 - e^{-2}(1+2) = 1 - 3e^{-2}$$

Question 13

M2L12 M2L13

Per the CLT, the approximate distribution of the sample mean $\bar{X}_n \sim \text{Nor}(\mu, \sigma^2/n)$

For a Bern(0.9), distribution $\mu = 0.9$ and $\sigma^2 = 0.9(1 - 0.9) = 0.09$

$\bar{X}_{100} \sim \text{Nor}(0.9, 0.09/100) \sim \text{Nor}(0.9, 0.0009)$

Question 14

M2L6 M3L8

Recall from the lecture

Example (Discrete Uniform): Consider a D.U. on $\{1, 2, \ldots, n\}$, i.e., $X = i$ with probability $1/n$ for $i = 1, 2, \ldots, n$. (Think of this as an n-sided dice toss for you Dungeons and Dragons fans.)

If $U \sim \text{Unif}(0, 1)$, we can obtain a D.U. random variate simply by setting $X = \lceil nU \rceil$, where $\lceil \cdot \rceil$ is the "ceiling" (or "round up") function.

The answer is the sum of two 10-sided dice.

Question 15

M1L8 M2L12 M3L8

Using properties of logarithms:

$$-3\ell n(U_1(1-U_2)U_3) = -3(\ell n(U_1) + \ell n(1-U_2) + \ell n(U_3))$$

$$= -3\ell n(U_1) + -3\ell n(1-U_2) + -3\ell n(U_3)$$

Recall that $-\frac{1}{\lambda}\ell n(1-U) \sim -\frac{1}{\lambda}\ell n(U) \sim \text{Exp}(\lambda)$

Each of these is an exponential random variable with $\lambda = 1/3$

Adding these results in an Erlang distribution with $n = 3$ and $\lambda = 1/3$

Question 16

M2L6 M2L12 M3L8

This is a direct result of the inverse transform theorem.

If X has distribution $\Phi(x)$ this means that X is a standard Normal random variable. The CDF for the standard Normal gets this special character of Φ since it is so special and widely used.

We set $\Phi(X) = U$ and solve for X by taking the inverse of each side.

This results in $X = \Phi^{-1}(U)$ which means that X will have a standard Normal distribution.

This is basically saying that if we substitute uniform random numbers between 0 and 1 into the inverse CDF of the standard Normal distribution, the result that is generated will be random variates from the standard Normal distribution.

Question 17

M2L5 M2L7 M3L8

The easiest way is to use the inverse transform theorem which states that the random variable $F(X) \sim \text{Unif}(0, 1)$. The expected value of a $\text{Unif}(0, 1)$ random variable is 0.5.

LOTUS can also be used with $h(X) = F(X)$.

If the integral shown in the solution document is confusing, then do something like u-substitution with $u = F(x)$ and $du = f(x)dx$.

Then for the limits, when $x = -\infty$ then $u = F(-\infty) = 0$ and when $x = \infty$ then $u = F(\infty) = 1$. (From the definition of a CDF)

This leaves us with the integral $\int_0^1 u du = \frac{u^2}{2}\Big|_0^1 = \frac{1}{2} - \frac{0}{2} = \frac{1}{2}$

Question 18

M1L8 M2L6 M3L8

$X_{i+1} = (5X_i + 1) \mod 8$ with $X_0 = 0$

To find U_{99}, we need to see what the cycle length of this LCG is.

$X_1 = (5X_0 + 1) \mod 8 = 1 \mod 8 = 1$
$X_2 = 6 \mod 8 = 6$
$X_3 = 31 \mod 8 = 7$
$X_4 = 36 \mod 8 = 4$
$X_5 = 21 \mod 8 = 5$
$X_6 = 26 \mod 8 = 2$
$X_7 = 11 \mod 8 = 3$
$X_8 = 16 \mod 8 = 0$

So, the LCG starts repeating after 8 iterations.

This implies that $X_{99} = X_{91} = \cdots = X_{11} = X_3 = 7$

A more direct way is to take $99 \mod 8 = 3$, so we use the same value as X_3.

Question 19

M2L6 M3L8

This is the “desert island” LCG and all of the statements are true!

Question 20

M4L3

Updating the FEL

What do we mean by “updating the FEL”?

Any time there’s an event, the simulation may update the chronological order of the FEL’s events by

- inserting new events,
- deleting events,
- moving them around, or
- doing nothing.

Question 21

M4L3

Every discrete-event simulation language maintains a FEL somewhere deep in its cold, cold heart.

Question 22

M4L3

Notes on FELs

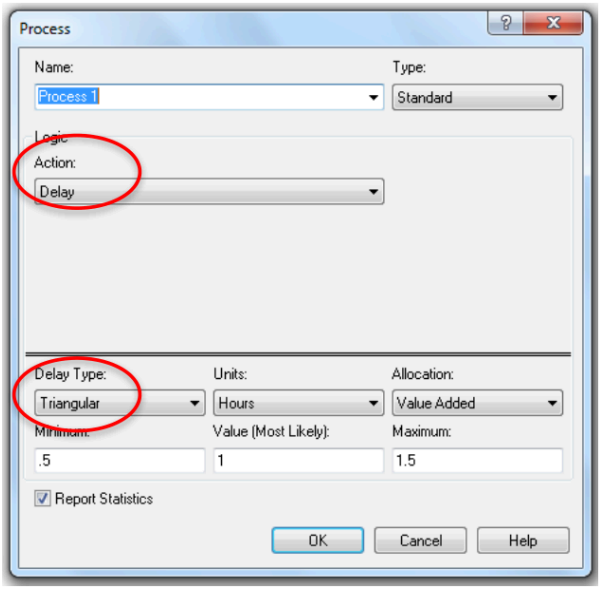
The system state can only change at event times. Nothing really happens between events.

The simulation progresses by sequentially executing (dealing with) the most imminent event on the FEL.

Question 23

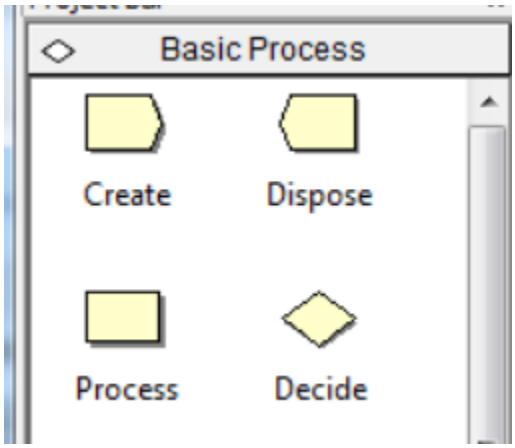
M5L5

- Process module has fields for
  - The name of the module
  - Type of action. This is where you can try to reserve a server or free a server who's currently in use. More on this in next lesson.
  - How long will you be delayed? I.e., what is the service time?
- Dispose module
  - Nothing deep here – you just get rid of entities.
  - You can name the module.



Question 24

M5L3 M5L8



Question 25

M3L1

f(x+h)	x	h	f(x)[1+h(x-3)]	
f(0)	0	-	-	2
f(0+0.01)	0	0.01	f(0)[1+(0.01)(0-3)]=2[1 - 0.03]	1.94
f(0.01+0.01)	0.01	0.01	f(0.01)[1 + (0.01)(0.01-3)]=1.94[1-0.0299]	1.881994

Question 26

M3L2

$$= \frac{1}{4}[(1 + e^{(0.42)^4}) + (1 + e^{(0.11)^4}) + (1 + e^{(0.73)^4}) + (1 + e^{(0.89)^4}))]$$

$$= \frac{1}{4}[2.032 + 2.000 + 2.328 + 2.873] = \frac{1}{4}(9.233) = 2.308$$

Question 27

M1L6 M3L4

This is a direct use of the method described in the modules and lessons referenced above.

Question 28

M2L7 M2L12

X/Y	0.2	0.4	0.6	0.8
0.2	$\frac{0.2}{0.2+0.2} = 0.5$	$\frac{0.2}{0.2+0.4} = 0.333$	$\frac{0.2}{0.4+0.6} = 0.25$	$\frac{0.2}{0.2+0.8} = 0.2$
0.4	$\frac{0.4}{0.4+0.2} = 0.667$	$\frac{0.4}{0.4+0.4} = 0.5$	$\frac{0.4}{0.4+0.6} = 0.4$	$\frac{0.4}{0.4+0.8} = 0.333$
0.6	$\frac{0.6}{0.6+0.2} = 0.75$	$\frac{0.6}{0.6+0.4} = 0.6$	$\frac{0.6}{0.6+0.6} = 0.5$	$\frac{0.6}{0.6+0.8} = 0.428$
0.8	$\frac{0.8}{0.8+0.2} = 0.8$	$\frac{0.8}{0.8+0.4} = 0.667$	$\frac{0.8}{0.8+0.6} = 0.57$	$\frac{0.8}{0.8+0.8} = 0.5$

The sum of all the cells will be 8. $8/16 = 0.5$

Question 29

M2L12 M3

The reason $U_i < p_i$ works is because of the uniformity of the uniform random variable. The interval is between 0 and 1. The probability that a uniform random number will fall in an interval of length p, e.g. [0, p] is p.

You can “prove” this by using the PDF of the uniform distribution.

$$P(0 < U < p) = \int_0^p 1dx = x\Big|_0^p = p$$

For the first shot the probability that we make it is 0.7; however, the random uniform number we generated is greater than this, so this means we missed.

Shot i	U_i	p_i	$U_i < p_i$?	Make/Miss ?
1	0.83	0.7	No	Miss
2	0.16			
3	0.95			
4	0.47			
5	0.37			
6	0.65			

Since we missed the first shot, our probability goes down to 0.6, but our uniform number is less than p, so we made the second shot.

Shot i	U_i	p_i	$U_i < p_i$?	Make/Miss ?
1	0.83	0.7	No	Miss
2	0.16	0.6	Yes	Make
3	0.95			
4	0.47			
5	0.37			
6	0.65			

Now we are feeling confident, so the probability goes up to 0.9. However, we got unlucky and missed the third shot!

Shot i	U_i	p_i	$U_i < p_i$?	Make/Miss ?
1	0.83	0.7	No	Miss
2	0.16	0.6	Yes	Make
3	0.95	0.9	No	Miss
4	0.47			
5	0.37			
6	0.65			

We continue this process until we make 4 shots.

Shot i	U_i	p_i	$U_i < p_i$?	Make/Miss ?
1	0.83	0.7	No	Miss
2	0.16	0.6	Yes	Make
3	0.95	0.9	No	Miss
4	0.47	0.6	Yes	Make
5	0.37			
6	0.65			

Shot i	U_i	p_i	$U_i < p_i$?	Make/Miss ?
1	0.83	0.7	No	Miss
2	0.16	0.6	Yes	Make
3	0.95	0.9	No	Miss
4	0.47	0.6	Yes	Make
5	0.37	0.9	Yes	Make
6	0.65			

Shot i	U_i	p_i	$U_i < p_i$?	Make/Miss ?
1	0.83	0.7	No	Miss
2	0.16	0.6	Yes	Make
3	0.95	0.9	No	Miss
4	0.47	0.6	Yes	Make
5	0.37	0.9	Yes	Make
6	0.65	0.9	Yes	Make

We made the 4th shot on our 6th attempt.

Question 30

M3L5

i	I_i	A_i	T_i	W_i^Q	S_i	D_i
Bill	8				3*6=18	
Tom	2				3*2=6	
Angie	5				3*3=9	
Ursula	2				3*1=3	

i	I_i	A_i	T_i	W_i^Q	S_i	D_i
Bill	8	8	8	0	18	26
Tom	2	10			6	
Angie	5	15			9	
Ursula	2	17			3	

Now we serve Angie next since she is the next in alphabetical order.

i	I_i	A_i	T_i	W_i^Q	S_i	D_i
Bill	8	8	8	0	18	26
Tom	2	10			6	
Angie	5	15	26	11	9	35
Ursula	2	17			3	

i	I_i	A_i	T_i	W_i^Q	S_i	D_i
Bill	8	8	8	0	18	26
Tom	2	10	35	25	6	41
Angie	5	15	26	11	9	35
Ursula	2	17			3	

i	I_i	A_i	T_i	W_i^Q	S_i	D_i
Bill	8	8	8	0	18	26
Tom	2	10	35	25	6	41
Angie	5	15	26	11	9	35
Ursula	2	17	41	24	3	44

The first customer leaves at time 26.

Question 31

M3L5

Another way of thinking about this is to look at the number of customers in the system in each interval.

[8, 10]: Bill (1)
[10, 15]: Bill and Tom (2)
[15, 17]: Bill, Tom, and Angie (3)
[17, 20]: Bill, Tom, Angie, and Ursula (4)

1(10 - 8) + 2(15 - 10) + 3(17 - 15) + 4(20 - 17)=2+10+6+12=30

Question 32

M3L5

Joey will make 2(6 + 2 + 3 + 1) = 24

Joey will lose 0.5(0 + 25 + 11 + 24) = 30

24 - 30 = -6

Question 33

On some of the practice exams, there were questions related to pop culture, which were considered bonus questions. Please do not stress over these, as they are meant to be fun and easy free points. If questions such as these appear on actual exams, you will get credit for them regardless.